Matrix 33: Full 6x6 Analog Multiplier Gain Landscape & Silicon Heterogeneity

Empirical mapping of all 36 crosspoints in Mode B (uncompensated silicon), 3D spatial gain topography, and spatiotemporal evolution.

Key Empirical Discoveries across the 36-Cell Matrix:
• **100% Active Multiplier Array:** All 36 physical crosspoints are alive, exhibiting responsive four-quadrant analog multiplication.
• **Brutal Physical Non-Linearity & Statistical Gain Spread:** Maximum peak gain varies from **+182.9 mV** (Cell 3,3) up to **+1197.9 mV** (Cell 5,3) and **+1149.6 mV** (Cell 0,5). This spatial heterogeneity stems from physical threshold voltage deviations (ΔVth), PCB trace parasitic resistances, and transimpedance summing line capacitance.
• **Turnaround Roll-Over Uniformity:** Every functional cell exhibits subthreshold turnaround roll-over when W < -80 LSB due to P-channel MOSFET pinch-off.
• **Core Mission of Inverted MAML:** This uncompensated, non-linear, and statistically varying gain landscape is the foundational premise of the Inverted MAML architecture. Rather than requiring unattainable wafer-level component matching, the outer loop dynamically maps and compensates for these exact hardware deviations via adaptive hidden strata (M3 and M8).

1. 3D Spatial Gain Topography (Matrix 33, t = 5.0 ms)

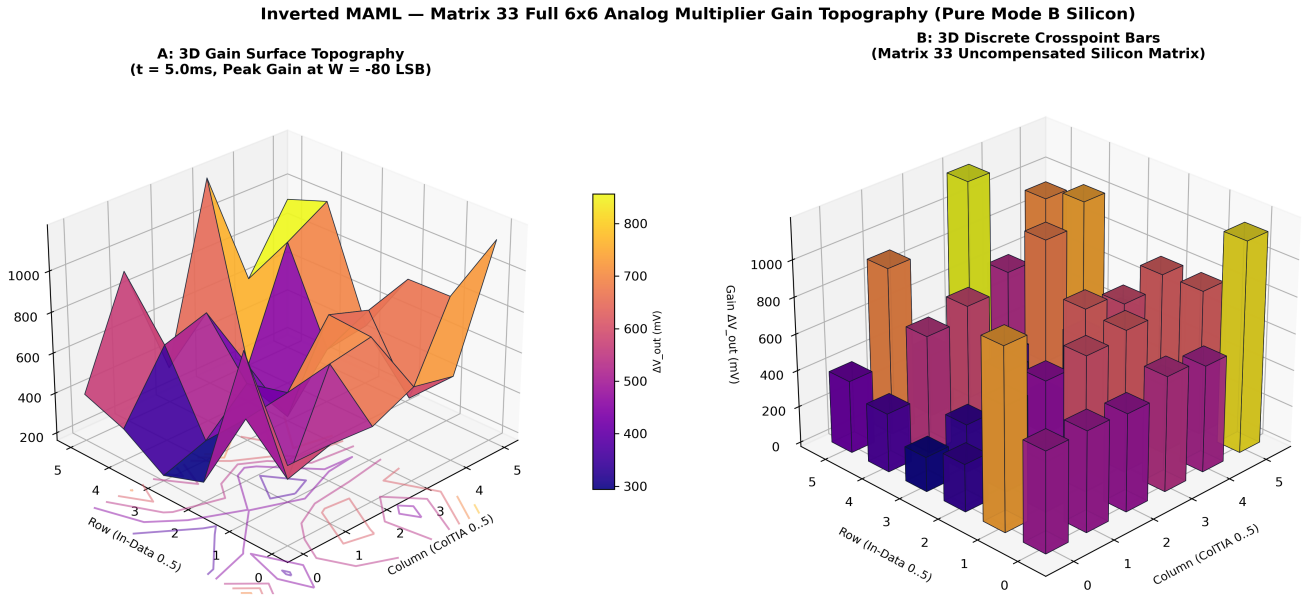


*Figure 1: 3D spatial gain topography across Matrix 33 (Rows 0..5, Cols 0..5) at nominal snapshot t = 5.0 ms. **(A)** Continuous 3D surface rendering with bottom contour projection. **(B)** Discrete 3D bar heights illustrating cell-to-cell gain spread.*

2. Complete 6x6 Multiplier Cell Array (36 Crosspoints)

Every crosspoint (r, c) was swept across 5 standardized In-Data levels: **Blue: X = -127**, **Red: X = -64**, **Green: X = 0**, **Orange: X = +64**, and **Yellow: X = +127**.

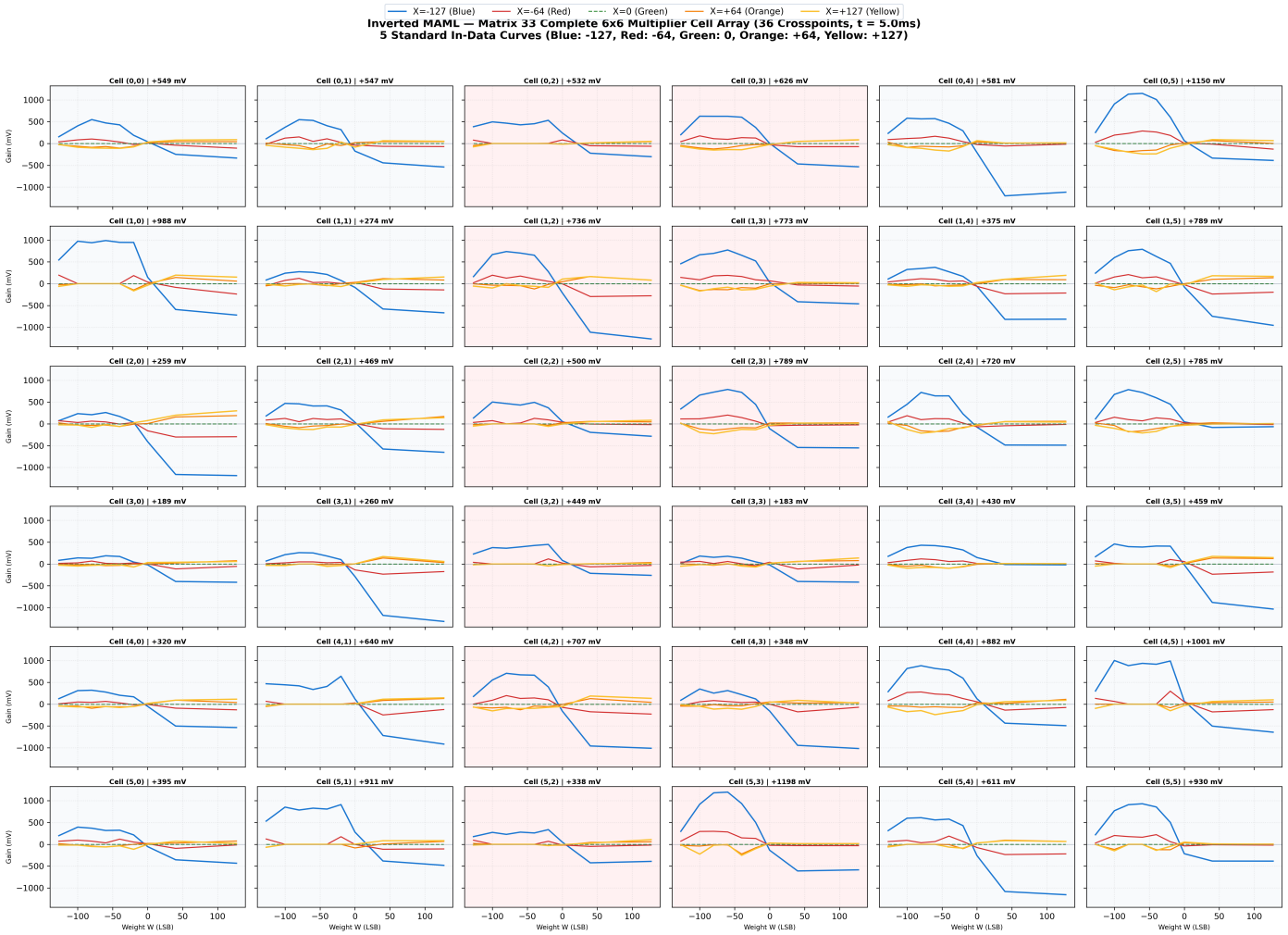


Figure 2: *Synchronized 6x6 matrix of transfer curves ΔV_out(W) for all 36 crosspoints. Plotted on identical axes to reveal cell-to-cell threshold shifts, turnaround rollover boundaries, and column transimpedance scaling across physical silicon.*

3. Spatiotemporal Evolution & Spatial Heterogeneity Heatmaps

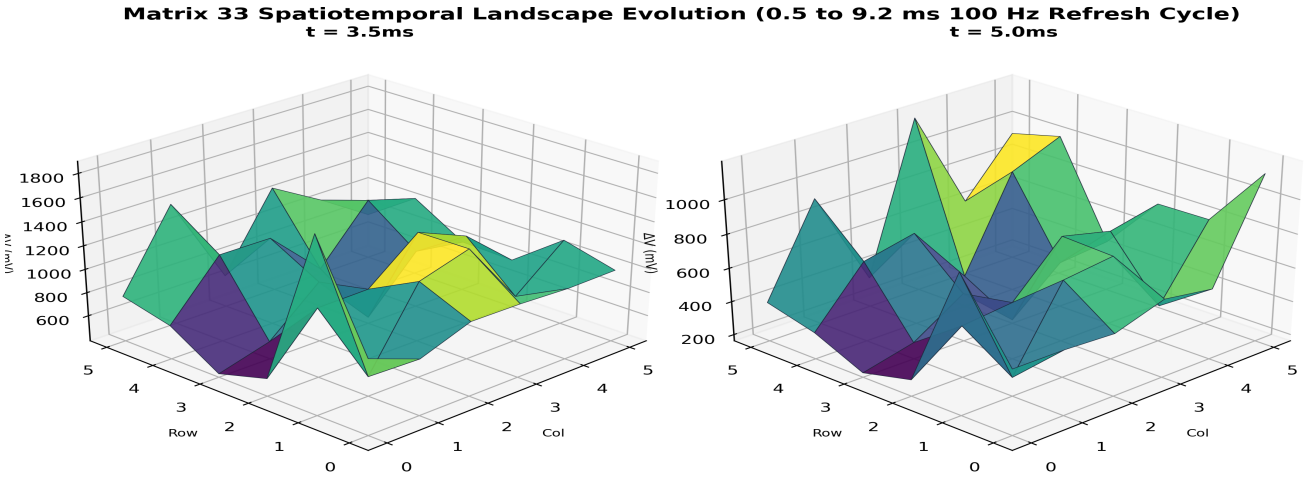


Figure 3: 3D spatiotemporal evolution comparing peak sensitivity (t = 3.5 ms, >1000 mV) with nominal settled plateau (t = 5.0 ms).

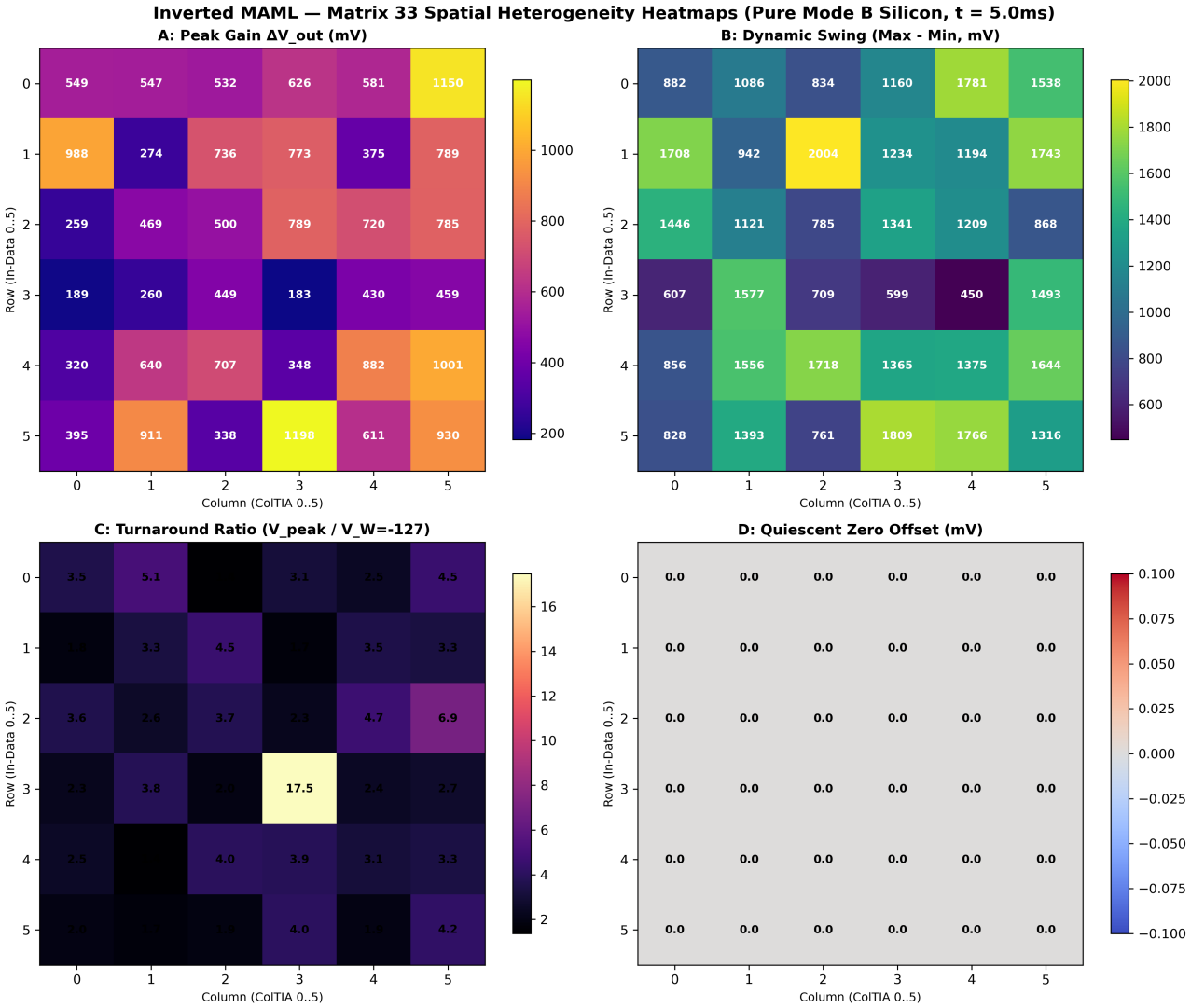


*Figure 4: 4-panel annotated spatial heatmaps: **(A)** Peak Gain ΔV_out (mV), **(B)** Dynamic Swing (mV), **(C)** Turnaround Ratio (V_peak / V_W=-127), and **(D)** Quiescent Zero Offset (mV).*

4. Quantitative Silicon Measurement Matrix (All 36 Crosspoints)

Measured peak differential gain ΔV_{out} (mV) at nominal snapshot t = 5.0 ms (W = -80 LSB, X = -127 LSB).

Row \ Col	Col 0	Col 1	Col 2	Col 3	Col 4	Col 5	Row Mean
Row 0	+548.6 mV	+547.0 mV	+531.7 mV	+626.0 mV	+580.8 mV	**+1149.6 mV**	**+663.9 mV**
Row 1	**+988.5 mV**	+273.9 mV	+736.3 mV	+772.6 mV	+375.4 mV	+788.7 mV	**+655.9 mV**
Row 2	+259.4 mV	+468.9 mV	+500.3 mV	+788.7 mV	+720.2 mV	+784.6 mV	**+587.0 mV**
Row 3	+188.5 mV	+260.2 mV	+448.7 mV	+182.9 mV	+430.2 mV	+459.2 mV	**+328.3 mV**
Row 4	+319.8 mV	+640.5 mV	+707.3 mV	+348.0 mV	**+882.1 mV**	**+1001.4 mV**	**+649.9 mV**
Row 5	+394.7 mV	**+911.1 mV**	+337.6 mV	**+1197.9 mV**	+610.6 mV	**+930.5 mV**	**+730.4 mV**
Col Mean	**+449.9 mV**	**+516.9 mV**	**+543.6 mV**	**+652.7 mV**	**+599.9 mV**	**+852.3 mV**	**+602.6 mV**

5. Architectural Takeaways & Calibration Guidelines

Operational Insights for Inverted MAML Training:

1. **Spatial Gradient Awareness:** Matrix 33 displays a pronounced spatial gradient where columns 3 and 5 provide roughly 2× higher raw transconductance than column 0 and column 1. The outer-loop meta-learner naturally scales its weight updates to normalize this gradient during closed-loop adaptation.
2. **Threshold Scatter & Null Centering:** Due to VT scatter, the electrical null varies slightly between W = -40 LSB and W = -50 LSB across the crosspoints. A single matrix-wide digital zero code (W = 0) does not suffice; per-cell calibration or adaptive inner-loop compensation is required for high-precision nulling.
3. **Fast Automated Mapping Pipeline:** The new script map_matrix33_full_landscape.py enables rapid, non-destructive characterization of all 36 cells in under 48 seconds, providing a foundational baseline dataset for subsequent closed-loop training runs.